

3.1.2 Electromagnetic Radiation and Quantum Phenomena

- **The photoelectric effect**

Work function ϕ , threshold frequency f_0 , photoelectric equation $hf = \phi + E_k$; the stopping potential experiment is not required.

- **Collisions of electrons with atoms**

The electron volt.

Ionisation and excitation; understanding of ionisation and excitation in the fluorescent tube.

- **Energy levels and photon emission**

Line spectra (e.g. of atomic hydrogen) as evidence of transitions between discrete energy levels in atoms.

$$hf = E_1 - E_2$$

- **Wave-particle duality**

Candidates should know that electron diffraction suggests the wave nature of particles and the photoelectric effect suggests the particle nature of electromagnetic waves; details of particular methods of particle diffraction are not expected.

$$\text{de Broglie wavelength } \lambda = \frac{h}{mv},$$

where mv is the momentum.